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RESEARCH-ARTICLE

What is a User Interface, again? A Survey of Definitions of User Interface: Our shared and implicit understanding of the concept of User Interface

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Research on user interface theory is my main project. Currently, with several partners at Francisco Marroquín university in Guatemala, we are working on designing experiences to learn from blind people and the way they use technology. Other projects include topics like programming languages for kids and soft skills development.

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Published: 16 November 2022

[Citation in BibTeX format](#)

MexIHC '22: Mexican International
Conference on Human-Computer
Interaction
November 16 - 18, 2022
Virtual Event, Mexico

What is a User Interface, again? A Survey of Definitions of User Interface

Our shared and implicit understanding of the concept of User Interface

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ABSTRACT

A central object of study in Human-Computer Interaction is the User Interface (UI). The number of websites, papers, books, videos, and articles that attempt to explain the concept, attests this. In this study a collection of definitions of User Interface was assembled and classified seeking to identify parallels. Several queries were used in Google, Bing, digital libraries, and other sources. With more than 130 entries collected, we found that most of them concentrate in the elements of physical-sensorial contact between humans and artifacts. We called them thin definitions because they overlook elements of processing in the side of the computer, and perception, thinking, and understanding in the human side. Thick definitions, that dig deeper into the concept, considering human and machine capabilities as part of the UI, are less numerous but important to study, advance, and promote. We will argue for the need of a new definition and propose one.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); HCI theory, concepts and models; Interaction design; Interaction design theory, concepts and paradigms; • **Software and its engineering** → Software organization and properties; Extra-functional properties; Software usability.

KEYWORDS

User Interface, Human-Computer Interaction, Design, Engineering, Research, Education, Definitions

ACM Reference Format:

Leonel, V, Morales Díaz. 2022. What is a User Interface, again? A Survey of Definitions of User Interface: Our shared and implicit understanding of the concept of User Interface. In *9th Mexican International Conference on Human-Computer Interaction (MexIHC '22), November 16–18, 2022, Virtual Event, Mexico*. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3565494.3565504>

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MexIHC '22, November 16–18, 2022, Virtual Event, Mexico

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ACM ISBN 978-1-4503-9904-3/22/11...\$15.00

<https://doi.org/10.1145/3565494.3565504>

1 INTRODUCTION

The most popular definition of user interface, by number of exact matches in the results of an online search query, is this:

“User interfaces are the access points where users interact with designs” [17]

At the moment of writing this paper it appears in at least 168 different web sites. Several features in this short text are prominent. It relies on the concept of access points to convey the idea of an encounter between the user and the device. Those access points would enable users to interact with something that has been designed. A designer with the task of designing a user interface may wonder which are the access points available to use, what kind of interactions are possible, and how the way the artifact was built may constraint the access points or the interactions.

The engineer who has to build the designed interface may wonder why a certain access point has been enabled while other has been blocked, whether a particular interaction makes sense, and why some functionalities of the device have been neglected and left without an access point.

What these reflections expose is that the text of that definition is of little use for the designer or the engineer even though their work is directly concerned with what it intends to define. It is not clear why this particular definition became so popular but, as will be shown in this paper, similar definitions are frequent. It is relatively easy to stumble upon definitions like that while researching underlying theories about user interfaces. Popular definitions present a general idea, most of the time too general to serve any purpose beyond eliciting an intuition of the concept.

In this research project a collection of definitions of the notion of user interface was assembled and their prevalence in the web was estimated. Each definition was analyzed and classified in a set of categories. By counting the items in each category and their occurrences on the web, it is possible to conjecture on the dominant type of definitions and how much the concept has been developed and even identify prominent problems with it at its current level. The purpose of the project is to signal the need for a more profound and applicable theory of user interfaces –often referred to as UI– starting by its very concept.

1.1 The Need for a Theory of User Interfaces

This study is not the first to notice the need for a theory of user interfaces. Lars-Erik Janlert and Erik Stolterman wrote in a 2015 paper “We believe that our field needs to have a better understanding of one of its core concepts, interface, because interaction and interactivity would seem to require an interface.” [7] Janlert and Stolterman identify four different thought styles about UIs: surface

of contact between matching objects, boundary of an independent object, means of controlling an object, and a means for impressions and expressions. These thought styles provide guidance and structure to search for a definition as the basis for a theory.

Although the interest of HCI practitioners and researchers has expanded to include every aspect of the user experience, the user interface remains an important subject of study. It makes sense to say that the experience of an interactive system of any kind, is heavily influenced by who is experiencing it and hence, the designer has to respect and work with that identity. To design an experience, or at least to attempt it, the designer must learn who is the user and be sure of that knowledge as much as possible. The UI, on the other hand, is also dependent on what the interactive product is, what functionalities it provides to work with specific objects, and what needs of the user it is designed to solve. The design of the user interface is much more controllable by the designer and because it has an important role in shaping the user's experience, it makes sense to dedicate an important amount of work to it. What this means is that the interest of HCI practitioners on the user experience has not relegated the importance of a theory of user interfaces. On the contrary, it has made it more relevant.

There are many theories that supplement and support the design and construction of user interfaces. Shneiderman et.al. groups theories by type and by human capacity [15]. In the first group they put descriptive, explanatory, prescriptive and predictive theories. In the second, motor, perceptual and cognitive ones. It is clear that these theories do a great service to UI design, but they are not theories of the UI. They are applicable in this field, and not for all cases, but for particular ones where the domain to which they belong coincides with the domain of application of the interactive system.

A theory about user interfaces should start with a good understanding of its very notion, a definition most theorists can agree. There is no such agreement for now. As this study will show, definitions abound, some are more popular than others, and most present epistemological problems that prevent their further refinement. In the book "The Structure of Scientific Revolutions", Thomas Kuhn says "proliferation of versions of a theory is a very usual symptom of crisis" [25], what we have in this case is proliferation at the basis of the theory.

The lack of a solid theory situates a field at the level of an art. In the case of UIs, this has been recognized by several authors, for example, Laurel [11] and Bennett and Flach [3]. In a paper published in 1981, T. P. Moran wrote about the UI: "it should be efficient to use and easy to learn; it should be consistent, logical, 'natural'; amen. There are at present no systematic methods or body of knowledge to help the designer accomplish these goals for the interface—it is a purely intuitive endeavor." [12] As pointed by Shneiderman et al. [15] there are several bodies of knowledge that help the designer but the lack of a "systematic method" is arguable still present.

Some have suggested that the focus should be on designing interactions rather than interfaces [2]. The works of Michael Beaudouin-Lafon and coauthors, proposing Instrumental Interaction as a new model [18] and the principles of reification, polymorphism and reuse [19] are of great value to look into that direction. In fact, HCI needs both, theories of interaction and theories of user interfaces. What we mean by interaction, the notion behind that word, is not unequivocal as shown in a study of 35 years of CHI publications

where Hornbæk et al., affirm that "The notion of interaction is essential to human-computer interaction, yet rarely studied." [6]. There are similar studies that show that the concept of user interface has been constantly addressed in HCI research, for example Jørgensen in [8] and Diaz and Chevez [4].

Instead of trying to make sense of the concept of UI by analyzing the literature as Hornbæk et al., in [6], in this study the focus was put on how the term is defined popularly and what are the features of the definitions that were found.

2 COLLECTING DEFINITIONS ONLINE

The collecting effort for this study started more than five years ago but was not systematized until 2019. A database for storing the definitions was set and a special Twitter account and hashtag were used to constantly publish the text of the definitions found including a screen capture of the webpage or PDF file where it was seen. This helped to quickly locate and compare definitions and keep an online journal of findings.

The idea was to see what an individual with Internet access would find if she queried "user interface" on standard search engines like Google, or special repositories like digital libraries, Google Scholar, video tutorials, Google Books, online encyclopedias, and others.

The notion of user interface is often described in domains like interactive design, user experience, web design, industrial design, robotics, artificial intelligence applications, and related. The purpose of the explanation attempt varies, most of the time it is for educational purposes, but it can also be for documentation of systems, for reference as in dictionaries, or supplementary concept needed in other domains. In many places the term was used without any introduction as if the authors assumed that it was common knowledge. Those that offer some explanation about the notion of UI sometimes do it loosely or embedding it into another idea like user interface design, interaction design, and interactive system. To avoid confounding concepts, we decided to accept only those cases where a clear indication was patent of the intention of defining what a user interface is.

The text of our queries often included explicit phrases like "can be defined as", "is defined as" and indicatives like "understood as", "is the". Several times the text of the definition did not immediately followed the term "user interface" so we reviewed the redaction, context, and surrounding text, to be sure it was an acceptable definition. In some websites the term "interface" was used as a synonym and a definition provided for it. Others made the distinction between "interface" and "user interface". Again, to decide whether a definition was suitable for our collection required careful review.

Along with the main defining text we stored the following fields for each entry: URL where it was found, date of last visit, author if available, name of the site or resource, URL of publication in our Twitter account, date of the tweet, and notes. Not all definitions come from academic publications. Blogs, magazines, wikis, websites of technology or design companies, and videos were also included. We assumed that if the formal citation was needed it would be easy to get it from the original resource whose URL was stored in our database.

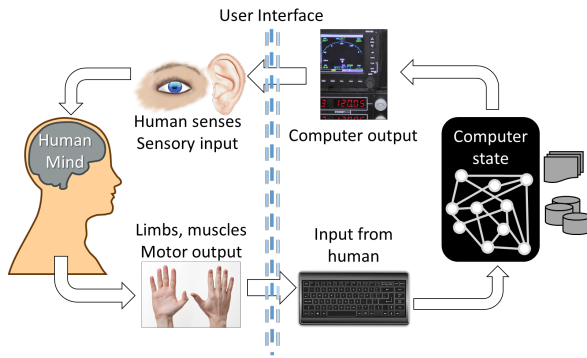


Figure 1: In HCI courses, it is common to illustrate the role of the User Interface with a diagram like this. The user, left side, is a human interacting with a computational artifact, right side. Arrows signal the input and output flow between them.

Additionally, a category was assigned to each entry in order to advance a classification. The basic categories were thin definitions and thick definitions. Thin definitions confine the user interface to the elements of physical contact between the user and the system, be it mechanical actuators like buttons or levers, or sensory stimulus like visual signals or sounds. Thick definitions extend the concept to include both human capabilities like sensation, perception, conceptualization, rationalization, analysis, and decision making, and computational resources like algorithmic processing, data storage, output preparation, and others.

The inspiration for naming the categories thin and thick came from a diagram we use in courses related to HCI. Figure 1 shows the diagram that illustrates the interaction enabled by the user interface. The UI is represented by a dividing wall that lets user output pass into the computer space to become input for it, and, conversely, it lets computer output to reach human senses and be perceived.

Thin definitions usually concentrate in describing the elements and operations that happen in the narrow space between the user and the computer. It is as if they were agnostic to the qualities and attributes of the beings on both sides. Thick definitions explicitly consider those features and make the user interface responsible of allowing their interconnectivity. Figure 2 shows this difference. Thin definitions, when taken seriously, liberate the designer or engineer from the duty of assuring that each side adapts correctly to the other. Thick definitions, because they pay attention to human and machine characteristics, let designers and engineers to address the problem of achieving satisfactory connection between them. In our survey, we found that some definitions were not purely thin or thick. This consideration led us to include variations. Depending on the emphasis made in the definition itself the category used could be a variant of thin or thick. The list of categories and subcategories used is presented in Table 1.

Regarding the date of publication of the resource containing the definition the criteria was not to exclude anyone based on antiquity. If the text could be found, then it was possible that a person wanting to learn about UIs would use it for any purpose and hence it was considered a valid entry.

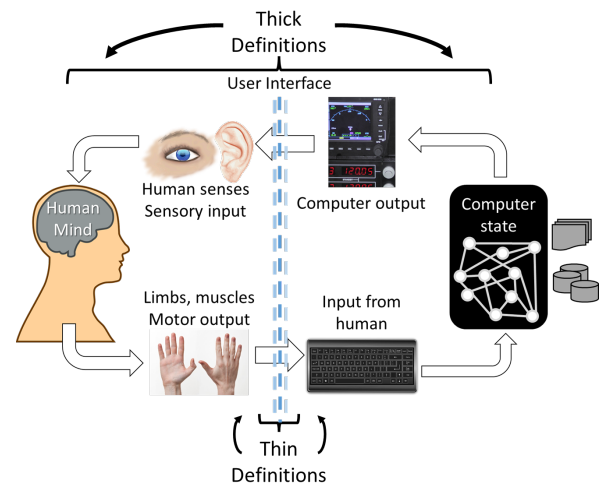


Figure 2: Thin and thick user interface definitions and the conceptual regions they cover. Thin definitions concentrate in the narrow space between the user and the system. Thick definitions widen the concept and include qualities and attributes of humans and computers.

More than 130 entries were collected, stored, tweeted, and categorized. The database keeps growing. The hashtag “#DefUI” (<https://twitter.com/hashtag/DefUI?f=live>) was used from a single handle, @IngSimple (<https://twitter.com/IngSimple>) for tweeting publicly the instances found. Figure 3 shows an example tweet. Storing results in a local database combined with posting them in Twitter, made it easier to search, review and avoid duplicate entries. The list of tweets can be accessed in: https://twitter.com/search?q=from%3AIngSimple%20DefUI&src=typed_query&f=live.

When the number of entries in the database was above 120, a series of queries were composed for each definition in order to find the approximate number of results that a search engine would report if queried with the exact definition text. As those texts could be lengthy and hence not acceptable in a search engine query, a special “core” version was composed for each one. The idea was to select the most representative phrase in the definition that was short enough but still fully indicative of a search match. That modification enabled the count of occurrences of each definition among the indexed pages of different search engines. Google, Bing, ACM’s Digital Library, and Google Scholar were tested. Google usually reported more matches than the others for most of the definitions. It seems that search engines try to prevent that pattern of use by indicating bogus or inflated numbers. The solution to that intentional ambiguity was to browse through all results pages until the last one, where a much more exact figure was reported. That is the number that got recorded.

With that addition the database now stores the definitions, with authors and sources, and the number of occurrences of each one.

3 RESULTS

Table 2 lists the results for the two general categories and for each of the sub-categories.

Table 1: Categories used for classification of entries in the definitions database

Category	Criteria of inclusion in this category
Thick	The definition includes both the users with their needs and operational capabilities, the computer or technology, and how to match them. Example: “User interfaces are a mapping from the sensory, cognitive, and social human world to these collections of functions exposed by a computer program.” [20]
Thick-Computer	Definitions that include the computer or technology as design considerations. The definition leans towards the computer. Example: “every computer’s user interface can be understood as state transitions caused by user actions.” [13]
Thick-User	Definitions that include the user, her goals, and abilities, and how the UI design has to approach them. The definition leans towards the user. Example: “a user interface consists of those parts of a system, which the user comes into contact with - physically, perceptually or conceptually” [9]
Thin	Definitions that comprise a narrow region of contact between user and computer and take little care of who the user is, what computational resources are available, and how to match both. Example: “A user interface is the method by which the user and the computer exchange information and instructions.” [21]
Thin-Ad-hoc	Thin definitions that emphasize the particular purpose of the system being built and hence are useful only for that design problem. Example: “A user interface is what managers see and use when they interact with a DSS” [14] DSS stands for Design Support System. This definition is valid only in the DSS domain.
Thin-Visual	Thin definitions that assume that the computer output is visual. They usually mention the screen or visible elements in the UI. Example: “An interface is the visible piece of a system that a user sees or hears or touches. Users come into contact with an interface when they use a system, often needing to get a task done.” [5]

Table 2: Summary of classification results.

Main category	Cases	%	Sub-categories	Cases	%
Thick	31	22%	Thick	6	4%
			Thick-Computer	11	8%
			Thick-User	14	10%
Thin	108	78%	Thin	79	57%
			Thin-Ad-hoc	16	12%
			Thin-Visual	13	9%
Totals	139	100%	Totals	139	100%

Of the 139 entries in the database 108 (78%) were in the thin category. Of those, 16 were ad-hoc definitions, produced for a particular purpose in the context of a specific project and hence assumed not to be intended for generalization. 13 were clearly referred to visual interfaces and thus had a limited scope because they exclude non-visual UIs. The remaining 79 were “pure” thin definitions.

In the side of the 31 thick definitions (22%) the majority, 14, leaned towards the user, meaning that they include considerations of human capabilities as part of the user interface. 11 leaned towards the computer and mention computational functions as integral of the UI. Only 6 were “pure” thick and include considerations of elements of both sides in the definition.

Besides the figures and examples, there are other insights and curiosities that we observed while collecting the data. Some thin definitions seem to have gained popularity during periods of time and were difficult to trace to their original proponent because the same text was found in several sites. We came across one case of debate in the literature, with one author analyzing and challenging the definition proposed by another. There could be other cases, but we did not encounter them.

In one case a form of circularity was manifest in the text: “Interface. A view or presentation of an object, service, or environment that a person (or group) interacts with, and the capabilities that provide for interaction across the interface.” [22]

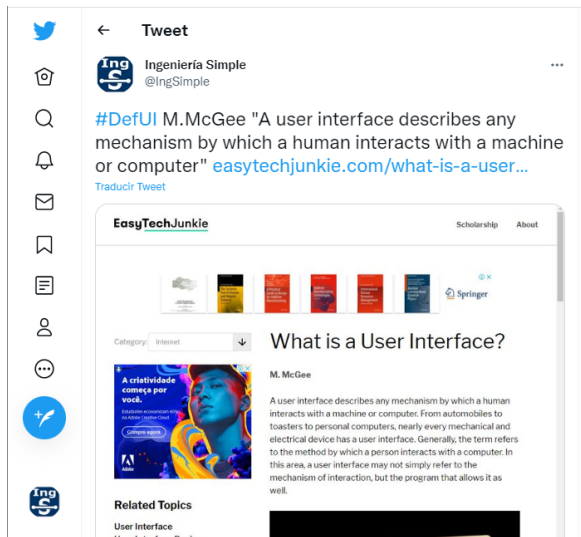


Figure 3: Each definition was shared publicly through a Twitter account. The tweet included the hashtag #DefUI, a portion of the text of the definition, the URL of the web page where it was found, a screen capture of that web page, and the author of it if available.

Definitions that explicitly reference the visual channel are not scarce. In fact, being the least abundant in the thin side, they are at least as abundant as any thick category. On the other hand, explicit mentions of sound, eye-tracking, voice driven or conversational, tactile, gesture based, brain-computer interaction, virtual reality, augmented reality, and other of the newest modes of interaction were almost not present.

3.1 Prevalence

Along with the one presented at the beginning of this paper, the most prevalent UI definitions are these: “in the industrial design field of human-computer interaction, is the space where interactions between humans and machines occur.” [23] 150 matches, and “Abbreviated UI, the junction between a user and a computer program. An interface is a set of commands or menus through which a user communicates with a program.” [16], 149 matches. The three of them are classified as thin definitions. In fact, this type is much more prevalent than the rest of categories.

The top representatives of each category are as follows:

- “User interfaces are the access points where users interact with designs” [17], Thin, 168 matches.
- “User Interface (UI) defines the way humans interact with the information systems. In Layman’s term, User Interface (UI) is a series of pages, screens, buttons, forms and other visual elements that are used to interact with the device” [24], Thin-Visual, 46 matches.
- “The user interface of a system consists of those aspects of the system that the user comes in contact with—physically, perceptually, or conceptually” [12], Thick-User, 30 matches.

- “An adaptive user interface is a software artifact that improves its ability to interact with a user by constructing a user model based on partial experience with that user” [10], Thick-Computer, 21 matches.
- “The user interface mediates between the core mechanics of the game and the player. It takes the challenges that are generated by the core mechanics (driving a racing car, for example) and turns them into graphics on the screen and sound from the speakers. It also turns the player’s button presses and movement on the keyboard or controller into actions within the context of the game” [1], Thin-Ad-hoc, 10 matches.
- “how the outer and the inner environments are mediated. Both the human and the technology also have invariant properties that together with the designed variant properties shape the process and outcomes of interaction. The final artefact reflects a designer’s implicit theory of this interplay” [13], Thick, 3 matches.

4 DISCUSSION

From the results obtained in the survey the most prevalent form of definition of the concept of user interface is thin. The availability of thick definitions indicates that there have been efforts to expand and complement the notion, but their variance implies lack of agreement which may be also a problem.

If availability translates into probability, then any person, designer, researcher, or student, seeking to understand the concept of user interface, in order to achieve a better design, understand and solve a problem more accurately, or learn about human-computer interaction, has more than 90% odds of finding a thin or thin-visual definition. If by chance, a thick definition is found, then a partiality towards the user or the computers is also likely to arise.

Thick definitions have to rely on models of the user and of the computer, in order to incorporate their elements into the definition. Because models are also a source of differences between concepts, this reliance may also add disparity in that class. In addition, understanding the subjacent model may be a burden for the reader, although we do not have a firm base for confirming that at this point.

4.1 Problems with thin definitions

From the results presented, thin definitions, because of their high prevalence, seem to be in a solid position. This means that a thin definition may be usually taken for valid without questioning or objection. Many of them share terms and get reinforced by repetition of similar enunciations. Their abundance suggests that there is no need to craft a new definition. Their lack of profundity is, nevertheless, a problem. If the person consulting the definition is looking for a quick description any of them will be good enough, but if she is looking for initial guidance for designing a new user interface (asking herself “what is what I have to design?”) or to evaluate, assess or compare user interfaces (“what makes this one better than others?”) or is trying to learn about human-computer interaction by looking at one of its fundamental components (“where do interaction occurs?”) then thin definitions will fall short.

Two problems are then apparent with thin definitions: lack of guidance ability and lack of a parametrical reference. The first problem, lack of guidance, occurs after learning a thin definition. Because they are too general and only superficially descriptive, they do not help set a goal of the type “[this] is what has to be [found / designed / evaluated / interacted with]”. The notion is learned but the person is still incapable of recognizing a UI for the purpose she might have. The definition did not guide her.

The lack of parametrical reference problem comes from the definition being unconnected to other concepts, or their parts and components not being described. If a person learns a UI definition that fails to situate the notion in a more general category and also fails to describe what are the features and traits that differentiates UIs from other concepts in that same general category, then there will be no reference to make practical use of the idea. The idea is described but no reference point was provided to reach it.

If the definition uses broad terms like “points of contact”, “surface of interaction” and similar ones, then practically anything in the interactive system or device matches the definition. The immediate conclusion is then that there is no reason to achieve more.

A better, and much needed, solution is to advance the development of thick definitions. At this moment, as our results show, they are scarce and lack a common theoretical ground to build the models they need to explain clearly and deeply what a user interface is.

5 A WORKING DEFINITION

Although this paper discusses the need for advancing the theory of user interfaces and shows evidence of the absence of a satisfactory definition of the concept that can be used as the basis for that advance, it is important to apply the experience obtained in this study to propose a new definition. Such definition should establish the artificial nature of user interface and hence start by stating that they are a kind of artifacts that require a guiding purpose and design work.

Then, it has to show the elements that directly relate to it and connect with it: a computer or interactive device with a set of capabilities like memory, storage, processing capacity, speed, input, and output possibilities. Under ideal conditions, where all computational power is available, these elements should not influence the design of a UI, but, because ideal conditions are never met and every design has to be finally implemented in a particular hardware, these considerations have to be included.

The human side of the interaction has different levels to address: motor, sensitive, perceptive, rational, emotional, cognitive, volitional. The goals of the user, both short and long-term, also have to be included.

Finally, because humans do not use computers just for the sake of using them, there must be some objects that are created, edited, handled, managed or informed about, with the assistance of the computer.

In summary, there are three elements interconnected through a user interface: a computational device, a (human) user, and a set of objects which are intentionally dealt within the process. With all those elements in mind, the working definition would be this:

“A user interface is an artifact, meaning that it has to be designed and built with a purpose, that connects a computational device and a human user, enabling the effective coupling of the motor, sensitive, perceptive, cognitive and volitional faculties of the human with the input, storage, processing, and output capacities of the device, for the set of tasks that are related to a particular set of objects.”

As a working definition, this one is pending validation and refinement. We need to assess it for clarity, comprehensiveness, minimality, and other desirable features like learnability and memorability. These are goals to be addressed in future work.

6 CONCLUSION

In this paper we have shown the results of a survey that searched online for definitions of “user interface”. The idea was inspired in the notions of interaction in the seminal works of several authors. There seemed to be more to user interfaces than the standard definition describes. We collected more than 130 texts from websites, online books, blogs, video tutorials, academic papers, and other sources, selecting them based on their proximity to a formal definition or the expressed intention in the text to define the term. The database also contains the number of matches that result from querying popular search engines with the text of the definitions, to obtain an estimation of the prevalence of each one.

Our data shows that thin definitions are the most common and the ones that share the common understanding of user interface, but we argue that they lack deepness and need to be replaced with thick definitions. Two problems were signaled with thin definitions: lack of guidance and lack of parametrical reference.

Currently available thick definitions have their own problems. They are scarce and need to rely on models that are dissimilar and sometimes difficult to understand. Our final conclusion is that the field of human-computer interaction needs to advance in the development and dissemination of a new and complete thick definition of “user interface”. To help advance in that direction a working definition of the concept of User Interface was proposed.

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